Pokhara University
Faculty of Science and Technology

Course Code: **CMP 357** | Full marks: 100
Course title: **Cyber Security** (3-1-2) | Pass marks: 45
Nature of the course: Theory & Practical | Total Periods: 45
Level: Bachelor | Program: BE

Course Description
This course provides an in-depth study of cybersecurity principles, including security threats, risk management, cryptographic techniques, network security, and ethical hacking. It explores web security and system protection mechanisms, enabling students to understand and mitigate cyber threats. The course combines theoretical concepts with hands-on practical work to equip students with real-world cybersecurity skills.

General Objectives
The course is designed with the following objectives:

- To understand fundamental concepts of cybersecurity, including threats, vulnerabilities, and risk management.
- To explore cryptographic principles and their application in securing data and communication.
- To analyze network security mechanisms such as firewalls, intrusion detection systems (IDS), and VPNs.
- To introduce ethical hacking and penetration testing.
- To learn about web security threats and protection strategies.
- To examine legal and ethical aspects of cybersecurity and data privacy.
- To provide hands-on experience with cybersecurity tools and techniques.

Methods of Instruction
The course will be delivered through a combination of:

- Lectures: Providing theoretical foundations of cybersecurity.
- Tutorials: Discussion-based learning and case studies.
- Practical Work: Hands-on exercises on cybersecurity tools, encryption, and ethical hacking.
- Quizzes & Assignments: Evaluating students' understanding of key concepts.
- Group Discussions & Presentations: Engaging students in real-world cybersecurity issues.
- Project Work: Implementing security solutions in real-world scenarios.

Contents in Detail with Specific Objectives

Specific Objective	Content
Students will understand the importance of cybersecurity, the challenges faced, key security principles (CIA), risk management frameworks, best practices in securing information systems, various types of malware and antivirus solutions.	**Unit 1: Introduction to Cybersecurity (7 hrs)** 1.1 Understanding cybersecurity: Definition, importance, and challenges. 1.2 Cyber threats, vulnerabilities, and attack vectors. 1.3 Security principles: Confidentiality, Integrity, and Availability (CIA). 1.4 Risk assessment and management frameworks. 1.5 Malware types: Viruses, worms,

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	ransomware, trojans. 1.6 Antivirus and endpoint security solutions.
Students will learn about cryptographic techniques, including symmetric/asymmetric encryption, cryptographic algorithms, hashing techniques, digital signatures, and secure communication protocols like SSL/TLS.	**Unit 2: Cryptography and Data Security(8 hrs)** 2.1 Introduction to cryptography: Symmetric vs. asymmetric encryption. 2.2 Cryptographic algorithms: DES, AES, RSA, ECC. 2.3 Hashing techniques: MD5, SHA-256. 2.4 Digital signatures and certificates. 2.5 Secure communication protocols: SSL/TLS.
Understanding of network security fundamentals, secure network design, VPNs, wireless security standards, and tools like Nmap and Wireshark for network scanning and penetration testing.	**Unit 3: Network Security (7 hrs)** 3.1 Network security fundamentals: Firewalls, IDS and IPS. 3.2 Secure network architecture and design. 3.3 VPNs and secure remote access. 3.4 Wireless security and encryption standards (WPA, WPA2, WPA3). 3.5 Network scanning and penetration testing tools (Nmap, Wireshark).
Students will Gain knowledge of ethical hacking methodologies, common attack techniques, the Metasploit framework, social engineering, and incident response in penetration testing.	**Unit 4: Ethical Hacking and Penetration Testing (7 hrs)** 4.1 Overview of ethical hacking and penetration testing. 4.2 Common hacking techniques: SQL injection, cross-site scripting (XSS), phishing. 4.3 Metasploit framework and exploit development. 4.4 Social engineering attacks and countermeasures. 4.5 Incident response and digital forensics.
Students will learn about web security vulnerabilities, secure coding practices, Web Application Firewalls (WAFs), secure authentication, session management, and lessons from major web security breaches.	**Unit 5: Web Security and Application Protection (8 hrs)** 5.1 Web security vulnerabilities: OWASP Top 10. 5.2 Secure coding practices for web applications. 5.3 Web application firewalls (WAF). 5.4 Secure authentication and session management. 5.5 Case studies of major web security breaches.
Students will learn about key cybersecurity laws, ethical issues, data privacy compliance, digital evidence handling, and case studies on law enforcement in cybersecurity.	**Unit 6: Cybersecurity Laws, Ethics, and Compliance (8 hrs)** 6.1Cyber laws and regulations (GDPR, NIST, ISO 27001, Electronic Transaction Act, National ICT Policy). 6.2Ethical issues in cybersecurity. 6.3Data privacy and compliance. 6.4Digital evidence and legal procedures.

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	6.5Case studies on cybersecurity law enforcement.

List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

1. Identifying security threats and vulnerabilities in real-world scenarios.
2. Implementing symmetric and asymmetric encryption techniques.
3. Understanding cryptographic hashing and digital signatures.
4. Configuring firewalls and network security tools.
5. Conducting penetration testing using Kali Linux.
6. Analyzing network traffic with Wireshark.
7. Performing web security assessments (SQL injection, XSS).
8. Setting up secure authentication mechanisms.
9. Configuring security policies in operating systems.
10. Understanding social engineering attacks through case studies.
11. Developing a secure web application.
12. Conducting an incident response exercise.
13. Studying cybersecurity compliance and legal issues.
14. Group discussion on emerging cybersecurity trends.

Practical Work

1. Hands-on cryptography: Implementing AES and RSA encryption.
2. Network security lab: Configuring firewalls, IDS, and VPNs.
3. Penetration testing: Using Kali Linux tools (Metasploit, Burp Suite).
4. Web security testing: Identifying vulnerabilities in web applications.
5. Forensic analysis: Extracting and analyzing digital evidence.
6. Configuring Linux and Windows security settings.
7. Wireshark lab: Analyzing packet captures.
8. Social engineering simulation: Awareness training on phishing and scams.
9. Project work: Developing a real-world cybersecurity solution.

Evaluation System and Students' Responsibilities

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Internal Evaluation (50 Marks)

Internal Evaluation

In addition to the formal end-semester exam(s), the internal (formative) evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation and presentation etc. The tabular presentation of the internal evaluation is as follows. The components may differ according to the nature of the subjects.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End Examination	50
Attendance and Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance and Class Participation	10%			

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Lab Report/ Project Report	20%			
Practical Exam/ Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50+50=100				

Students' Responsibility

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the semester-end examination. Failing to get such a score will be equated with NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End- Semester examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. Failure of a student to attend a formal exam, quiz, test, etc. won't qualify him/her for re-exam. ***Students are required to complete all the requirements defined for the completion of the course***

Prescribed Books and References

Prescribed Books

1. Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short, *Cybersecurity Essentials*, 1st Edition, Wiley
2. William Stallings, *Cryptography and Network Security: Principles and Practice*, 7th Edition, Pearson
3. Michael T. Simpson, Kent Backman, James Corley, *Hands-On Ethical Hacking and Network Defense*, 3rd Edition, Cengage Learning
4. Michael Sikorski, Andrew Honig, *Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software*, 1st Edition, No Starch Press

References

1. Michael E. Whitman, Herbert J. Mattord, *Principles of Information Security*, 7th Edition, Cengage Learning
2. Raef Meeuwisse, *Cybersecurity for Beginners*, Updated Edition (2023), Cyber Simplicity Ltd
3. Bruce Schneier, *Applied Cryptography: Protocols, Algorithms, and Source Code in C*, 2nd Edition, Wiley
4. Kevin D. Mitnick, William L. Simon, *The Art of Deception: Controlling the Human Element of Security*, 1st Edition, Wiley
5. Dafydd Stuttard, Marcus Pinto, *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws*, 2nd Edition, Wiley
6. Jonathan Clough, *Principles of Cybercrime*, 2nd Edition, Cambridge University Press
7. OWASP Web Security Testing Guide (OWASP.org).

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